

BB84 protocol simulator

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Abstract

"Ugh, yet another infodump on Quantum Key Distribution! We don't need this!"

Maybe you don't, but I do. So, here is the BB84 protocol explained simply (I hope!). The purpose of these pages is to show a simple simulation of the BB84 protocol using the Qiskit library developed by IBM (all info on the site: [IBM Qiskit \[sita\]](#)); verifying that indeed, albeit in an ideal scenario, the acquisition of information about the key by an attacker induces an error on the key itself. This leads to a practical verification that the security of a quantum channel is not provided by computationally hard-to-calculate functions, but by the quantum mechanics properties of the channel. For the theoretical proof, please refer to [\[SP00\]](#) and [\[Mur\]](#). Keeping in mind that QKD, to date, remains one of the few applications of quantum mechanics in the cryptographic field that we can actually touch firsthand, I wish you a good read!

1 BB84: what it is and how it works

BB84 [\[Cha84\]](#) is historically the first cryptographic protocol based on the properties of quantum mechanics. It owes its name to its inventors: Charles H. Bennett and Gilles Brassard, who conceived it in 1984. This protocol relies on two important features of quantum mechanics:

- 1) The superposition (see definition [1.1](#)) in which any particle obeying the laws of quantum mechanics finds itself breaks irreparably like a soap bubble the exact moment it is measured.
- 2) It is not possible to clone without errors (i.e., exactly duplicate) a quantum state that does not belong to a set of mutually orthogonal states (see definition [1.2](#)) known a priori. This result goes by the name of the *No-cloning theorem*, and wouldn't you know, we find ourselves in the exact situation to apply it.

Definition 1.1 (Superposition) *Superposition is the property by which a quantum system exists in a linear combination of multiple possible states prior to measurement. In essence, it defines the set of potential values the particle can take and the probability distribution associated with each outcome upon observation.*

Definition 1.2 (Orthogonality) *Two vectors $|v\rangle$ and $|W\rangle$ (that is, two bases) are orthogonal if their inner product is 0.*

Let's proceed to illustrate how the algorithm works. As always, the two main characters are Alice and Bob, who want to use a classical (hence insecure) channel to communicate. They therefore decide to exchange a key to encrypt the messages transiting on the authenticated classical channel, and talk without Eve, who is always lurking, being able to read their messages. The two also have a quantum channel at their disposal, which they use to agree on this key. It is important to note that even though the channel they use to exchange the key is a quantum channel, the agreed-upon key will be a "classical" key, made of bits 0 and 1. To do this, they choose the BB84 protocol.

But what does a quantum channel look like?

We can imagine a quantum channel as any physical medium that allows the transmission of particles that obey the laws of quantum mechanics. In our case, we can imagine photons (i.e., unitary packets of light) racing along a fiber optic cable.

Now back to the algorithm. First of all, Alice and Bob agree on polarization bases (NON-ORTHOGONAL), which will be used to polarize the photons sent inside the quantum channel. They can agree on the



Figure 1: The barber example.

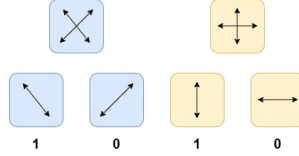


Figure 2: BB84 protocol bases shared by Alice and Bob.

bases using the classical channel, even if insecure: this information does not help Eve in any way. Those who are not familiar with quantum mechanics (not that I'm a luminary!) will be wondering: what does it mean to polarize a photon?

Well, that's easily said. Let's imagine that every photon is enveloped in a chaotic cloud. This cloud is nothing more than the photon's electric field. When a photon is polarized, that is, when it's passed through an oriented filter (we can imagine it as an actual slit through which the photon is squeezed), we force this electric field to oscillate only along the direction defined by the filter. Even more simply: let's imagine the photon has a messy bush of curls on its head (the electric field) and to fix its hair it decides to go to the barber. However, this barber is a peculiar guy and only has electric clippers capable of cutting hair at a certain angle θ . These electric clippers are the polarizing filters. Every time the photon goes to the barber, it comes home with a mohawk at an angle θ (see figure 1).

Let's compose ourselves. Alice and Bob choose two bases: a rectilinear, or computational one (oriented like standard Cartesian axes), and a diagonal or Hadamard one (tilted at 45° relative to the previous one). A bit value is associated with the polarizations of these bases. Figure 2 shows the bases and the bit values corresponding to the polarizations. Specifically: the diagonal bases, in light blue, are tilted at 135° and 45° , and correspond to the value 1 and 0 respectively; similarly, the vertical polarization (90°) corresponds to the bit value 1 and the horizontal one (0°) to the bit value 0.

Once the bases and their respective polarizations are shared, the classical channel is temporarily abandoned and we switch to the quantum one. Alice uses the agreed bases to send polarized photons to Bob through the quantum channel. For his part, Bob has no way of knowing which bases Alice used, and can only measure the photons he receives by choosing the bases randomly. What happens if Bob uses a different basis to measure than the one used by Alice to polarize the sent photon? In this case, the photon will be polarized with a 50% probability along one of the two directions of the basis chosen by Bob, as shown in figure 3. It's good to note that even using the wrong basis, Bob could still measure the bit value correctly, but as we'll see shortly, these "uncertain" bits won't be part of the shared key.

Figure 4 shows an example of the photon transmission procedure from Alice to Bob, described previ-

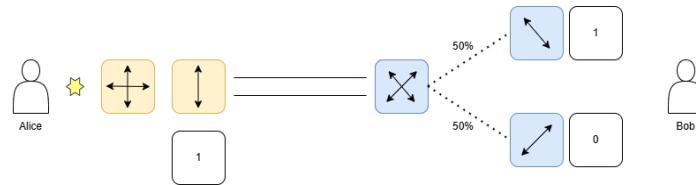


Figure 3: When Bob chooses the wrong measurement basis, the photon assumes one of the two polarizations of the basis chosen by Bob with a 50% probability.

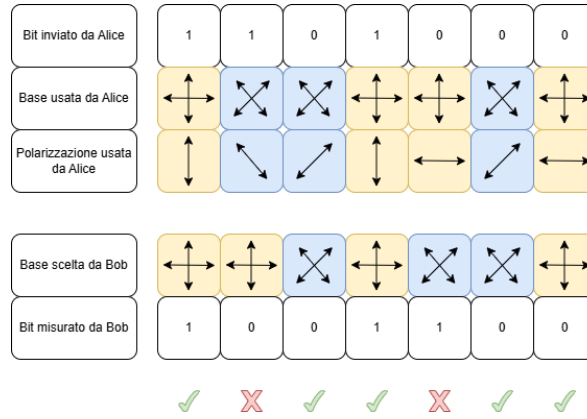


Figure 4: Sending photons from Alice to Bob and measurement.

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Once the transmission of the photons is complete, Alice and Bob use the classical channel to compare the measurement bases they used, keeping only the bits for which the bases match, while the others are discarded. The first k bits of the key are also compared (and then discarded as well): if they match, Alice and Bob haven't been intercepted; if there are errors, on the other hand, the sending of the photons has been compromised and must be repeated. The remaining bits constitute the shared key.

2 The role of the attacker

And Eve? How can she act? How do Alice and Bob figure out if the sending of photons on the quantum channel has somehow been intercepted?

Let's proceed in order. First of all, let's suppose Eve acts as a *Man - In - The - Middle*, inserting herself into the channel and measuring the passing photons, exactly like Bob does. To answer the third question, however, the features we stated at the beginning of section 1 come to our rescue: on the one hand, the first point guarantees that by performing the measurement, Eve irreversibly determines the value of the quantum particle; and the No-cloning Theorem ensures that Eve cannot introduce into the channel a photon with the same superposition as the polarized photon originally sent by Alice.

Okay, but if Eve is very lucky, she could choose for every photon the same basis Alice used to polarize! True, but if n photons are sent, the probability of Eve always choosing the correct basis is equal to $(1/4)^n$, which decreases as n grows. So, if Eve manages to guess n correct bases in a row by picking them at random, if I were her I'd go play a couple of lottery numbers. In addition, regardless of the basis chosen by Eve, the probability that no error is observed on the bit checked by Alice and Bob is equal (for n sent photons) to $(3/4)^n$. This value too, albeit less quickly than $(1/4)^n$, decreases as n increases.

NOTE: At the end of the previous section, it is reported that Alice and Bob, as the last step of the protocol, compare $k < n$ bits. In that case, the probability that no error is observed is equal to $(3/4)^k$. It is therefore important to choose k so that Eve is detected and at the same time the shared key has a suitable length to ensure the security of the messages exchanged on the classical channel.

3 Implementation of BB84 protocol

As already mentioned in the summary, the Python library used to implement the BB84 protocol is [Qiskit](#). The code is inspired by the article [\[sitb\]](#), and traces exactly the steps described in section 1:

1. The number of qubits that Alice sends to Bob is defined.
2. Alice randomly chooses the qubits to send and for each qubit the basis with which to polarize it.
3. Bob also chooses the bases with which he will make the measurements.

4. At this point, you can choose whether to introduce a Man-in-the-Middle (Eve) into the quantum channel or not.
5. A quantum circuit is then created (with the command `QuantumCircuit(qbit, bit)`), onto which the qubits polarized by Alice are sent: if the basis used to polarize is Z (the computational one), bit 0 remains unchanged and bit 1 is flipped (i.e., the NOT gate is applied); whereas if the basis is X (Hadamard basis), then with bit 0 you can directly use the Hadamard gate, while with bit 1 you first have to flip it then apply Hadamard. The reason why all these gates are applied is easily said: because of how the library is built, by default the qubit passing on the circuit is polarized as $|0\rangle$. Therefore, using the computational basis, bit 0 is already correctly polarized, whereas to have bit 1 with the same basis it is necessary to use a NOT gate, which "rotates" the polarization of the qubit by 90° . Similarly, for bit 0 it is possible to apply the Hadamard gate directly, while for bit 1 it is necessary to add the NOT gate.
6. Once the qubits are sent, it's measurement time. It's important to observe that both Eve and Bob perform the same operations on the circuit: if previously it was chosen to insert an attacker in the channel, then the piece of code dedicated to Eve will also be executed, otherwise, we skip directly to Bob's part. Therefore Bob (and Eve), based on the bases they chose, perform the measurements and store the results.
7. Once the part relative to the quantum channel is completed, we move to the classical channel where Alice and Bob compare the bases used. During the comparison, the bits for which the basis does not match are discarded.
8. A certain number of bits are then selected (in this case 40% of the bits remaining from the previous step) which Alice and Bob reveal and then discard. Thanks to this comparison, they manage to understand if the exchange of qubits suffered interference by Eve: if there are no errors in the revealed bits, the channel remained private and the key is secure; otherwise, it will be necessary to discard the key and start over with the protocol.

4 Observations

In this chapter, we will discuss some points worthy of attention regarding the implementation of the BB84 protocol. First of all: the simulation presupposes an ideal channel, with no possibility of errors, from which it follows that the QBER (Quantum Bit Error Rate, i.e., the percentage of incorrect bits received out of the total transmitted) is 0 if there are no attackers interfering on the channel, and different from 0 otherwise. In reality, this is not the case: there are losses and distortions caused by the physical medium that introduce errors making the QBER different from 0 even in the absence of attackers. Therefore, a "reconciliation" is necessary, meaning an error threshold is set for the bits compared at the end of the protocol, in the classical channel: if the QBER exceeds this threshold, the communication on the quantum circuit is considered compromised and the protocol is aborted; otherwise, algorithms are used to correct the errors without revealing further bits of the key. The error threshold for the "reconciliation" phase is set based on the medium and experience.

Another fairly important assumption made in the simulation is that the classical channel is authenticated, meaning Alice and Bob are sure they are talking to each other and there are no people pretending to be one of the two interlocutors. Furthermore, the classical channel is also error-free.

In short, this simulation lives in an ideal world where there are no errors other than those introduced by an attacker, who is therefore easily identifiable. The real world is obviously more complicated than this, but for now, we'll settle for understanding how the protocol works in theory.

5 Conclusions

Well, the purpose of these pages was to deeply understand how the BB84 protocol works, both from a theoretical but mostly practical point of view, and to get familiar with the Qiskit library. This library proved to be an extremely useful tool, and the [IMB Quantum Learning](#) site offers free courses to get familiar with both the library and quantum computing in general. Clearly, the goal of these

pages is entirely personal and does not have the presumption of positioning itself as research, let alone innovation, which it clearly isn't. I therefore declare the goal fully achieved (at least by me).

References

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